Chemical, Biological, Radiological and Nuclear (CBRN) Defense Research Institute
Ministry of National Defense, Seoul, Republic of Korea
2 November 2026

Dear Editors,

We submit *No Resolvable Agreement Between Descriptor Distance and Colorimetric Array Response: A Reanalysis of Sixteen Authentic Chemical Warfare Agents* for consideration as an Article.

What the paper is about, and why we bring it here. Chemical warfare agent detection is almost always developed and qualified on simulants, because the agents themselves are available to few laboratories. Every such qualification rests on one assumption: that a compound chosen to resemble an agent on molecular descriptors will resemble it in what the detector measures. We tested that assumption directly, against colorimetric response measured on sixteen authentic agents spanning nerve, blood, choking, vesicant and A-series compounds. This journal sets the standard for whether a detection claim has been demonstrated against the analyte it names. The question we ask is one step earlier and bears on every simulant-qualified device in the literature: whether the surrogate analyte was the right one to begin with.

Three counts state the result. In the three runs we reanalyse, one designated simulant was measured beside the agents, in one run of one array, and we found no second co-measurement in the literature we searched. Twelve exact Mantel tests and six Fisher combinations of them leave no agreement between descriptor distance and response distance below $q = 0.05$, against a design that resolves a rank correlation of 0.67 at 80% power where the largest observed is 0.57. On the thirty-five-compound panel, no agent of eleven has a valid match to an incumbent simulant, at a Benjamini–Hochberg adjusted value of 0.6299, and the compounds that displace the incumbents are themselves literature-proposed simulants. The array discriminates: in all six run-channel cells every compound agrees with itself across concentrations more closely than with the other compounds in that cell. What fails is the transfer from descriptor space to response space.

The specific case will interest sensor developers. Dimethyl methylphosphonate is the standard G-agent simulant and the benchmark compound for decontamination efficiency. Measured beside the agents in the ultraviolet channel, its response pattern correlates at $+0.76$ with tabun and $+0.65$ with sarin but at -0.49 with soman, -0.72 with cyclosarin and -0.70 with VX. A compound whose measured response is positively correlated with two G agents and negatively with the other two is not reproducing the series it stands in for. We draw no conclusion about its suitability for decontamination testing, which is the measurement it was designated against and which we did not perform.

What this paper is not, stated plainly. It reports no new sensor, no demonstration in a complex matrix, and no correlation against an established analytical method, and we recognise that your guidelines ask for those of a sensing paper. This is not a sensing paper. It is a measurement-science audit of the surrogate on which sensing papers are qualified, and the data it audits could not be produced any other way: the agents cannot be put into a complex environmental matrix outside a designated laboratory, and the comparison it makes is between two descriptions of the same compounds rather than between a sensor and a reference method. We would rather you weigh that against your scope now than have it surface in review.

Why we think it is nonetheless in scope. Two precedents in *ACS Sensors* bear on that judgement, and we would rather name them than leave them to be found. The first is our own. In 2025 this journal published a bioelectronic nose tested on authentic sarin and VX by a group that includes two of the present authors, and that paper noted in its abstract that detection research built on structural simulants may not carry over to the agents (*ACS Sens.* 2025, 10, 185–195). It asserted the risk and worked around it. The present manuscript

quantifies it, and we have cited that paper in the Introduction and positioned our contribution against it explicitly, so that the novelty on offer is the measurement and not the claim. The second is that this journal publishes work asking whether the components a sensing paper depends on have in fact been validated, as in the recent practical guide to choosing and validating aptamers, which opens on a reproducibility problem in reagents rather than on a device (*ACS Sens.* 2026, DOI 10.1021/acssensors.6c02357), which the manuscript cites in its Introduction. Simulants are the analyte-side equivalent of that problem. Every simulant-qualified detection claim in the literature rests on an unmeasured transfer, and we think the readership that builds those detectors is the readership that should see the measurement.

Data provenance. The colorimetric response data are reanalysed from three earlier studies by this group, in *Journal of Hazardous Materials*, *Discover Sensors* and *Sensors and Actuators B*, all cited in the manuscript. Those papers reported classification performance for three separate agent sets. None of them compared response to molecular descriptors, none placed the three sets in one analysis, and no figure, table or passage is reproduced here. The descriptor calculations, the permutation tests and every result in this manuscript are new.

What we propose instead of a replacement compound. We propose none, because selecting one on descriptor similarity would repeat the error the paper is about. What the manuscript specifies is the measurement that would settle the question: a bridging run in which candidate simulants and the agents they stand in for are exposed on the same plate, in the same optical channels, at the same concentrations, within one session. That run yields a response-space distance between a simulant and its agent measured under identical conditions, which no amount of descriptor work can supply. It needs no new chemistry and no new instrument.

Reporting and availability. All agents were produced and handled at an Organisation for the Prohibition of Chemical Weapons designated laboratory under Chemical Weapons Convention procedures. Analysis code, descriptor tables, parsed regulatory values with their sources and every figure script are deposited in a public repository cited in the manuscript, and all permutation designation and Mantel tests are exhaustive rather than sampled, with the two sampled controls declared and seeded, so that every number reported can be reproduced from the deposit. The manuscript has not been published elsewhere and is not under consideration at another journal. The authors declare no competing financial interest.

Yours sincerely,

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